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Course: **CSE103 (Structured Programming)**

Lab: **08**

Topics Today

1. 2D Array/ Matrix

Problem 1

Write a C program to add two matrices.

Sample Input	Sample Output
Enter the number of rows and columns: 2 2 Enter elements of the first matrix: 1 2 3 4 Enter elements of the second matrix: 5 6 7 8	Sum of the matrices: 6 8 10 12

Problem 2

Write a C program to subtract two matrices.

Sample Input	Sample Output
Enter the number of rows and columns: 2 2 Enter elements of the first matrix: 5 6 7 8 Enter elements of the second matrix: 1 2 3 4	Subtraction of the matrices: 4 4 4 4

Problem 4

Write a C program to perform Scalar matrix multiplication.

Sample Input	Sample Output
Enter the number of rows and columns: 2 2 Enter the scalar value: 2 Enter elements of the matrix: 1 2 3 4	Scalar matrix multiplication: 2 4 6 8

Problem 5

Write a C program to multiply two matrices.

Sample Input	Sample Output
Enter rows and columns of the first matrix: 2 2 Enter elements of the first matrix: 1 2 3 4 Enter rows and columns of the second matrix: 2 2 Enter elements of the second matrix: 5 6 7 8	Multiplication of the matrices: 19 22 43 50

Problem 6

Write a C program to check whether two matrices are equal or not.

Sample Input	Sample Output
Enter the number of rows and columns: 2 2 Enter elements of the first matrix: 1 2 3 4 Enter elements of the second matrix: 1 2 3 4	Matrices are equal.

Problem 7

Write a C program to find sum of main diagonal elements of a matrix.

Sample Input	Sample Output
Enter the number of rows and columns: 3 3 Enter elements of the matrix: 1 2 3 4 5 6 7 8 9	Sum of main diagonal elements: 15

Problem 8

Write a C program to find sum of minor diagonal elements of a matrix.

Sample Input	Sample Output
Enter the number of rows and columns: 3 3 Enter elements of the matrix: 1 2 3 4 5 6 7 8 9	Sum of minor diagonal elements: 12

Problem 9

Write a C program to find sum of each row and column of a matrix.

Sample Input	Sample Output
Enter the number of rows and columns: 2 3 Enter elements of the matrix: 1 2 3 4 5 6	Sum of each row: 6 15 Sum of each column: 5 7 9

Problem 10

Write a C program to interchange diagonals of a matrix.

Sample Input	Sample Output
Enter the number of rows and columns: 3 3 Enter elements of the matrix: 1 2 3 4 5 6 7 8 9	Matrix after interchanging diagonals: 3 2 1 4 5 6 9 8 7

Problem 11

Write a C program to check if a matrix is upper triangular.

Sample Input	Sample Output
Enter the number of rows and columns: 3 3 Enter elements of the matrix: 1 2 3 0 4 5 0 0 6	The matrix is upper triangular.

Problem 12

Write a C program to check if a matrix is lower triangular.

Sample Input	Sample Output
Enter the number of rows and columns: 3 3 Enter elements of the matrix: 1 0 0 2 3 0 4 5 6	The matrix is lower triangular.

Problem 13

Write a C program to find the sum of upper triangular matrix.

Sample Input	Sample Output
Enter the number of rows and columns: 3 3 Enter elements of the matrix: 1 2 3 0 4 5 0 0 6	Sum of upper triangular matrix: 21